

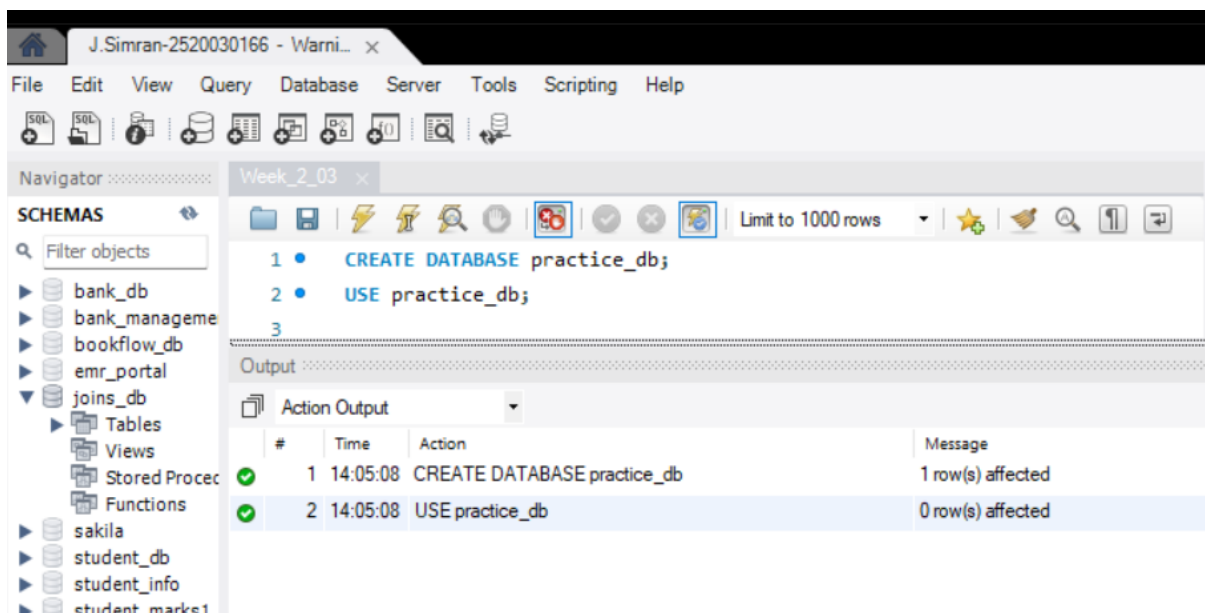
WEEK-2

(example problems)

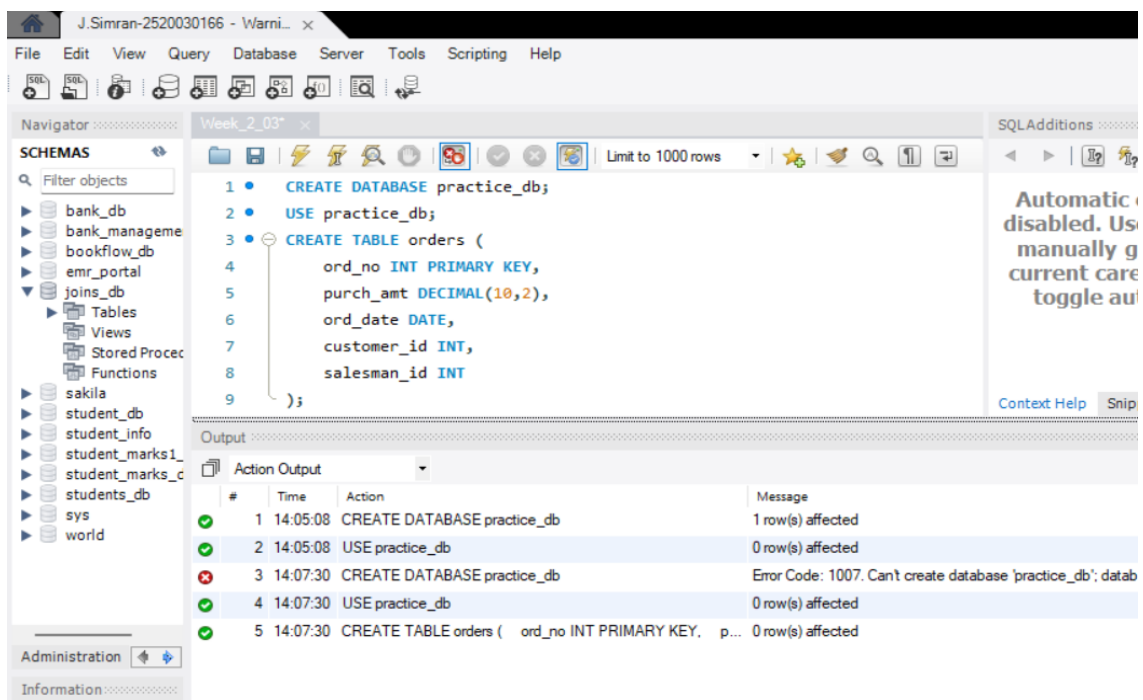
J.Simran

2520030166

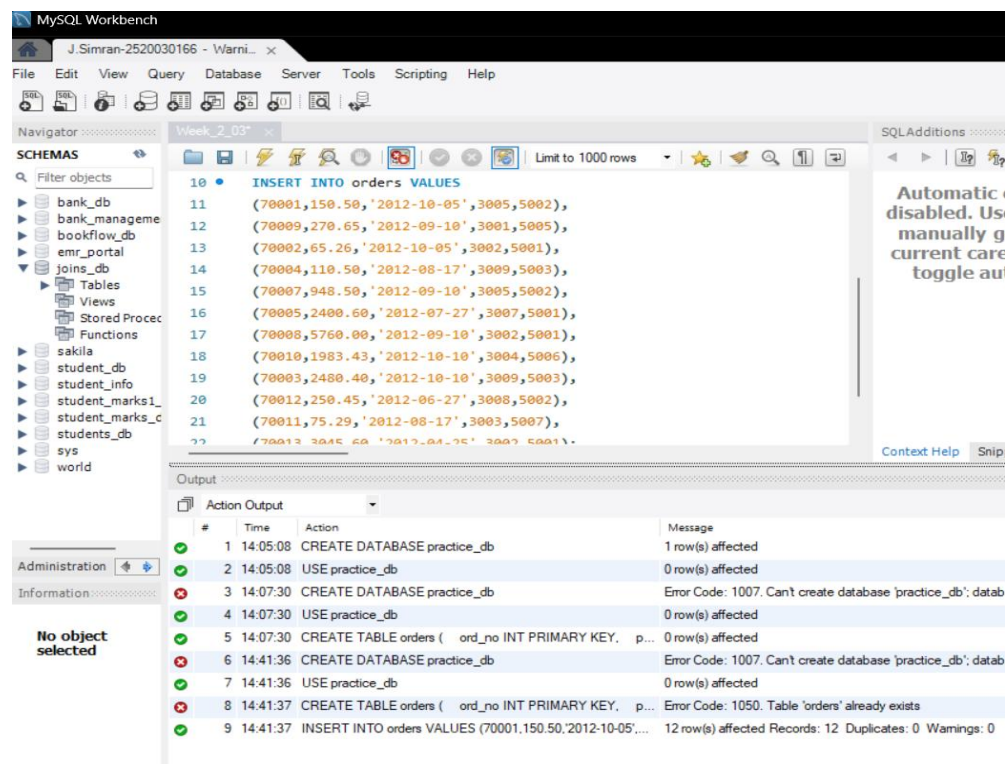
Step 1: Create a New Database & Select the Database



Step 2: Create the Table



Step 3: Insert Sample Data



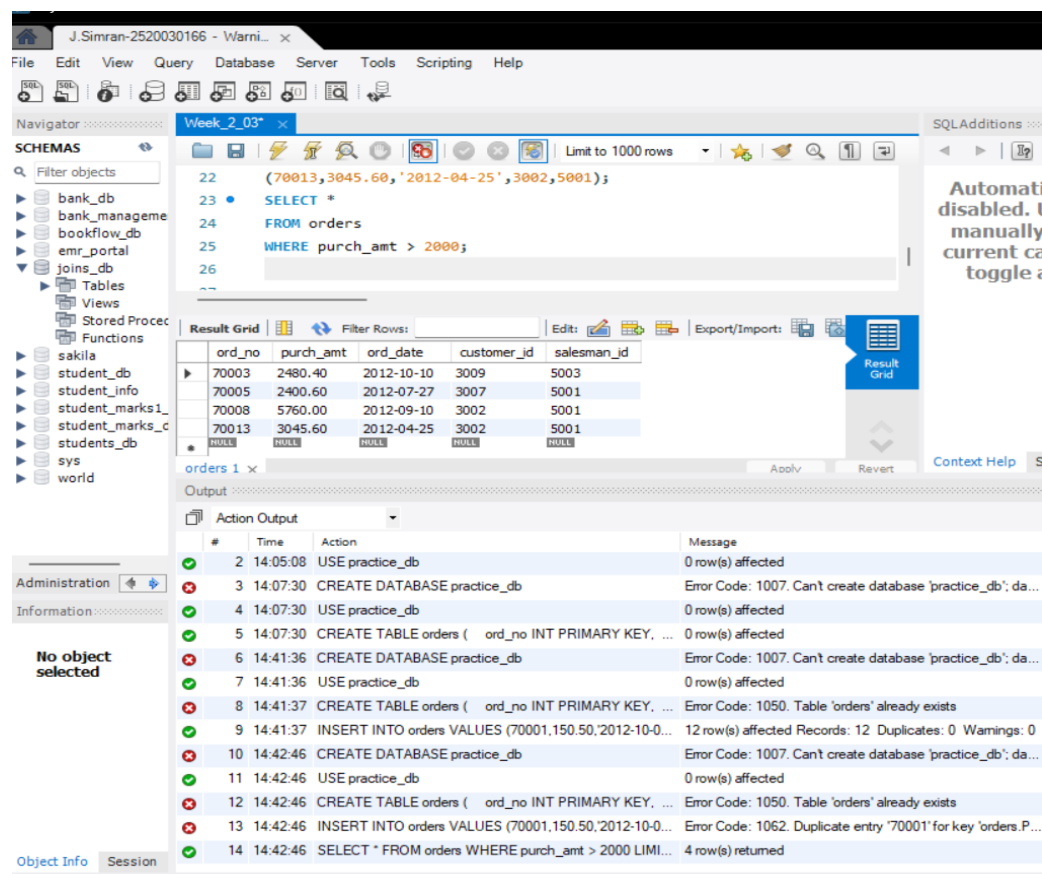
MySQL Workbench interface showing the execution of an SQL query. The query is:

```
INSERT INTO orders VALUES  
(70001,150.50,'2012-10-05',3005,5002),  
(70009,270.65,'2012-09-10',3001,5005),  
(70002,65.26,'2012-10-05',3002,5001),  
(70004,110.50,'2012-08-17',3009,5003),  
(70007,948.50,'2012-09-10',3005,5002),  
(70005,2400.60,'2012-07-27',3007,5001),  
(70008,5760.00,'2012-09-10',3002,5001),  
(70010,1983.43,'2012-10-10',3004,5006),  
(70003,2480.40,'2012-10-10',3009,5003),  
(70012,250.45,'2012-06-27',3006,5002),  
(70011,75.29,'2012-08-17',3003,5007),  
(70013,3045.60,'2012-04-25',3002,5001);
```

The output pane shows the results of the query execution:

#	Time	Action	Message
1	14:05:08	CREATE DATABASE practice_db	1 row(s) affected
2	14:05:08	USE practice_db	0 row(s) affected
3	14:07:30	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; datab...
4	14:07:30	USE practice_db	0 row(s) affected
5	14:07:30	CREATE TABLE orders (ord_no INT PRIMARY KEY, p...	0 row(s) affected
6	14:41:36	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; datab...
7	14:41:36	USE practice_db	0 row(s) affected
8	14:41:37	CREATE TABLE orders (ord_no INT PRIMARY KEY, p...	Error Code: 1050. Table 'orders' already exists
9	14:41:37	INSERT INTO orders VALUES (70001,150.50,'2012-10-05'...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0

Step 4: WHERE Clause



MySQL Workbench interface showing the execution of an SQL query. The query is:

```
SELECT *  
FROM orders  
WHERE purch_amt > 2000;
```

The output pane shows the results of the query execution:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001

The output pane also shows the results of the query execution:

#	Time	Action	Message
2	14:05:08	USE practice_db	0 row(s) affected
3	14:07:30	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
4	14:07:30	USE practice_db	0 row(s) affected
5	14:07:30	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	0 row(s) affected
6	14:41:36	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
7	14:41:36	USE practice_db	0 row(s) affected
8	14:41:37	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
9	14:41:37	INSERT INTO orders VALUES (70001,150.50,'2012-10-05'...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0
10	14:42:46	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
11	14:42:46	USE practice_db	0 row(s) affected
12	14:42:46	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
13	14:42:46	INSERT INTO orders VALUES (70001,150.50,'2012-10-05'...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
14	14:42:46	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned

Step 5: ORDER BY

The screenshot shows the SQL Developer interface with a query window titled 'Week_2_03'. The query is as follows:

```
25 WHERE purch_amt > 2000;  
26 SELECT *  
27 FROM orders  
28 ORDER BY purch_amt DESC;  
29
```

The 'Result Grid' displays the following data:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006

The 'Action Output' pane shows the execution log with the following messages:

#	Time	Action	Message
8	14:41:37	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
9	14:41:37	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0
10	14:42:46	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
11	14:42:46	USE practice_db	0 row(s) affected
12	14:42:46	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
13	14:42:46	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
14	14:42:46	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
15	14:43:36	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
16	14:43:36	USE practice_db	0 row(s) affected
17	14:43:36	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
18	14:43:36	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
19	14:43:36	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
20	14:43:37	SELECT * FROM orders ORDER BY purch_amt DESC LI...	12 row(s) returned

Step 6: Aggregate Functions

The screenshot shows the SQL Developer interface with a query window titled 'Week_2_03'. The query is as follows:

```
27 FROM orders  
28 ORDER BY purch_amt DESC;  
29 SELECT SUM(purch_amt) AS total_revenue  
30 FROM orders;  
31
```

The 'Result Grid' displays the following data:

total_revenue
17541.18

The 'Action Output' pane shows the execution log with the following messages:

#	Time	Action	Message
15	14:43:36	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
16	14:43:36	USE practice_db	0 row(s) affected
17	14:43:36	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
18	14:43:36	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
19	14:43:36	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
20	14:43:37	SELECT * FROM orders ORDER BY purch_amt DESC LI...	12 row(s) returned
21	14:44:18	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
22	14:44:18	USE practice_db	0 row(s) affected
23	14:44:18	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
24	14:44:18	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
25	14:44:18	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
26	14:44:18	SELECT * FROM orders ORDER BY purch_amt DESC LI...	12 row(s) returned
27	14:44:18	SELECT SUM(purch_amt) AS total_revenue FROM order...	1 row(s) returned

Step 7: GROUP BY

The screenshot shows the SQL Developer interface with a query window titled 'Week_2_03'. The query is as follows:

```
29 SELECT SUM(purch_amt) AS total_revenue
30 FROM orders;
31 SELECT salesman_id,
32        SUM(purch_amt) AS total_sales
33 FROM orders
34 GROUP BY salesman_id;
```

The 'Result Grid' shows the results of the second query, grouped by salesman_id:

salesman_id	total_sales
5002	1349.45
5001	11271.46
5003	2590.90
5005	270.65
5006	1983.43
5007	75.29

The 'Output' window shows the execution log with the following messages:

#	Time	Action	Message
23	14:44:18	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
24	14:44:18	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
25	14:44:18	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
26	14:44:18	SELECT * FROM orders ORDER BY purch_amt DESC LI...	12 row(s) returned
27	14:44:18	SELECT SUM(purch_amt) AS total_revenue FROM order...	1 row(s) returned
28	14:44:52	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
29	14:44:52	USE practice_db	0 row(s) affected
30	14:44:52	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
31	14:44:52	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
32	14:44:52	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
33	14:44:52	SELECT * FROM orders ORDER BY purch_amt DESC LI...	12 row(s) returned

Step 8: HAVING

The screenshot shows the SQL Developer interface with a query window titled 'Week_2_03'. The query is as follows:

```
35 SELECT salesman_id,
36        SUM(purch_amt) AS total_sales
37 FROM orders
38 GROUP BY salesman_id
39 HAVING SUM(purch_amt) > 3000;
40
```

The 'Result Grid' shows the results of the query, filtered by the HAVING clause:

salesman_id	total_sales
5001	11271.46

The 'Output' window shows the execution log with the following messages:

#	Time	Action	Message
35	14:44:52	SELECT salesman_id, SUM(purch_amt) AS total_sal...	6 row(s) returned
36	14:45:48	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
37	14:45:48	USE practice_db	0 row(s) affected
38	14:45:48	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
39	14:45:48	INSERT INTO orders VALUES (70001,150.50,2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
40	14:45:48	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
41	14:45:48	SELECT * FROM orders ORDER BY purch_amt DESC LI...	12 row(s) returned
42	14:45:48	SELECT SUM(purch_amt) AS total_revenue FROM order...	1 row(s) returned
43	14:45:48	SELECT salesman_id, SUM(purch_amt) AS total_sal...	6 row(s) returned
44	14:45:48	SELECT salesman_id, SUM(purch_amt) AS total_sal...	1 row(s) returned

Step 9: Combine Clauses

MySQL Workbench

J.Simran-2520030166 - Warni...

File Edit View Query Database Server Tools Scripting Help

Navigator: Week_2_03*

SCHEMAS

Filter objects

- bank_db
- bank_manageme
- bookflow_db
- emr_portal
- joins_db
 - Tables
 - Views
 - Stored Proce
 - Functions
- sakila
- student_db
- student_info
- student_marks1
- student_marks_c
- students_db
- sys
- world

SQL Editor

```
41 SUM(purch_amt) AS total_purchase
42 FROM orders
43 GROUP BY customer_id
44 HAVING SUM(purch_amt) > 1000
45 ORDER BY total_purchase DESC;
46
```

Limit to 1000 rows

Result Grid

customer_id	total_purchase
3002	8870.86
3009	2590.90
3007	2400.60
3004	1983.43
3005	1099.00

Export: Wrap Cell Content

Automat disabled. manually current c toggle.

orders 16 orders 17 Result 18 Result 19 Result 20 Result 21 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
45	14:46:38	CREATE DATABASE practice_db	Error Code: 1007. Can't create database 'practice_db'; da...
46	14:46:38	USE practice_db	0 row(s) affected
47	14:46:38	CREATE TABLE orders (ord_no INT PRIMARY KEY, ...	Error Code: 1050. Table 'orders' already exists
48	14:46:38	INSERT INTO orders VALUES (70001,150.50,'2012-10-0...	Error Code: 1062. Duplicate entry '70001' for key 'orders.P...
49	14:46:38	SELECT * FROM orders WHERE purch_amt > 2000 LIM...	4 row(s) returned
50	14:46:38	SELECT * FROM orders ORDER BY purch_amt DESC LI...	12 row(s) returned
51	14:46:38	SELECT SUM(purch_amt) AS total_revenue FROM order...	1 row(s) returned
52	14:46:38	SELECT salesman_id, SUM(purch_amt) AS total_sal...	6 row(s) returned
53	14:46:38	SELECT salesman_id, SUM(purch_amt) AS total_sal...	1 row(s) returned
54	14:46:38	SELECT customer_id, SUM(purch_amt) AS total_pur...	5 row(s) returned

Object Info Session

No object selected